

Integration — Lesson 35 Test: Volumes II — Off-Axis & Cross-Sections

10 questions · show your setup and work in the box · exact answers (fractions and π), no decimals · "set up" means write the integral but do NOT evaluate

Name: _____ Date: _____ Score: _____ / 10

1. (a) The curve $y = \sqrt{x}$ is revolved about the line $y = -3$. What is the radius of the disk at position x ? (b) The same curve is revolved about $y = 6$ (the curve stays below that line). Now what is the radius? 1 pt

2. The region between the line $y = x$ and the line $y = -1$, for $0 \leq x \leq 3$, is revolved about $y = -1$. Find the volume. 1 pt

3. The region between $y = \sqrt{x}$ and the line $y = -2$, for $0 \leq x \leq 1$, is revolved about $y = -2$. Find the volume. (Expand the square carefully.) 1 pt

4. The region bounded by $x = y^2$, the y-axis, and the line $y = 3$ is revolved about the y-axis. 1 pt
Find the volume.

5. The base of a solid is the region between $y = x$ and the x-axis on $[0, 3]$. Cross-sections perpendicular to the x-axis are squares. Find the volume. 1 pt

6. The base of a solid is the region between $y = \sqrt{x}$ and the x-axis on $[0, 2]$. Cross-sections perpendicular to the x-axis are semicircles with diameter on the base. Find the volume. 1 pt

7. The region bounded by $y = x$, the line $y = 3$, and the y-axis is revolved about the line $y = 3$. 1 pt
Find the volume, then check your answer with the cone formula $V = \frac{1}{3}\pi r^2 h$.

8. The region between $y = \sqrt{x}$ and the x-axis on $[0, 4]$ is revolved about the line $y = -2$. SET UP the volume integral (washers — do not evaluate). 1 pt

9. The base of a solid is the region between the line $y = 3$ and the line $y = x$, for $0 \leq x \leq 3$. Cross-sections perpendicular to the x-axis are equilateral triangles. Find the volume. 1 pt

10. Two solids share the same base. Solid A has square cross-sections and volume 10. Solid B has semicircle cross-sections (diameter on the base). (a) Find the volume of Solid B. (b) Explain in one sentence why Solid A's answer has no π in it. 1 pt
